

Cramer's Rule

Cramer's rule is needed in a variety of theoretical calculations. For instance, it can be used to study how the solution of $A\vec{x} = \vec{b}$ is affected by changes in the entries of $\vec{b}$.

However, the formula is inefficient for hand calculations, except for 2×2 or perhaps 3×3 matrices.

For any $n \times n$ matrix A and any $\vec{b}$ in $\mathbb{R}^n$, let $A_i(\vec{b})$ be the matrix obtained from A by replacing column i by the vector $\vec{b}$.

$$A_i(\vec{b}) = [\vec{a}_1 \dots \underset{\substack{\uparrow \\ \text{col } i}}{\vec{b}} \dots \vec{a}_n]$$

Theorem 7 - Cramer's Rule

Let A be an invertible $n \times n$ matrix. For any $\vec{b}$ in $\mathbb{R}^n$, the unique solution $\vec{x}$ of $A\vec{x} = \vec{b}$ has entries given by

$$x_i = \frac{\det A_i(\vec{b})}{\det A}, \quad i = 1, 2, \dots, n \quad (1)$$

Gabriel Cramer

Geneva

3 July 1704 - 4 January 1752

Proof

Denote the columns of A by $\vec{a}_1, \dots, \vec{a}_n$ and the columns of the $n \times n$ identity matrix I by $\vec{e}_1, \dots, \vec{e}_n$. If $A\vec{x} = \vec{b}$, the definition of matrix multiplication shows that

$$\begin{aligned} A(I_i(\vec{x})) &= A[\vec{e}_1 \dots \vec{x} \dots \vec{e}_n] \\ &= [A\vec{e}_1 \dots A\vec{x} \dots A\vec{e}_n] \\ &= [\vec{a}_1 \dots \vec{b} \dots \vec{a}_n] = A_i(\vec{b}) \end{aligned}$$

By the multiplicative property of determinants,

$$(\det A)(\det I_i(\vec{x})) = \det A_i(\vec{b})$$

The second determinant on the left is simply x_i . (Make a cofactor expansion along the i th row). Hence,

$(\det A)x_i = \det A_i(\vec{b})$. This proves (1) because A is invertible and $\det A \neq 0$.

3.3 Cramers rule, Volume and Linear Transformations

II

Example 1

Use Cramer's rule to solve the system

$$3x_1 - 2x_2 = 6$$

$$-5x_1 + 4x_2 = 8$$

Solution

View the system as $A\vec{x} = \vec{b}$. Using the notation introduced above,

$$A = \begin{bmatrix} 3 & -2 \\ -5 & 4 \end{bmatrix}, \quad A_1(\vec{b}) = \begin{bmatrix} 6 & -2 \\ 8 & 4 \end{bmatrix}, \quad A_2(\vec{b}) = \begin{bmatrix} 3 & 6 \\ -5 & 8 \end{bmatrix}$$

Since $\det A = 2$, the system has a unique solution.

[this follows from:

For a square matrix A , if $\det A \neq 0$, means the matrix is invertible. And if A is invertible, then $A\vec{x} = \vec{b}$ has exactly one solution

By Cramer's rule,

$$x_1 = \frac{\det A_1(\vec{b})}{\det A} = \frac{24 + 16}{2} = 20$$

$$x_2 = \frac{\det A_2(\vec{b})}{\det A} = \frac{24 + 30}{2} = 27$$

3.3 Cramer's rule, Volume and Linear Transformations

III

Application to Engineering

A number of important engineering problems, particularly in electrical engineering and control theory, can be analyzed by Laplace transforms. This approach converts an appropriate system of linear differential equations into a system of linear algebraic equations whose coefficients involve a parameter. The next example illustrates the type of algebraic system that may arise.

Example 2

Consider the following system in which s is an unspecified parameter. Determine the values of s for which the system has a unique solution, and use Cramer's rule to describe the solution.

$$\begin{aligned} 3s x_1 - 2x_2 &= 4 \\ -6x_1 + sx_2 &= 1 \end{aligned}$$

Solution

View the system as $A\vec{x} = \vec{b}$, then

$$\underbrace{\begin{aligned} 3s x_1 - 2x_2 &= 4 \\ -6x_1 + sx_2 &= 1 \end{aligned}}_{A\vec{x}} = \underbrace{\begin{aligned} 4 \\ 1 \end{aligned}}_{\vec{b}} \quad A = \begin{bmatrix} 3s & -2 \\ -6 & s \end{bmatrix}$$

$$A_1(\vec{b}) = \begin{bmatrix} 4 & -2 \\ 1 & s \end{bmatrix}, \quad A_2(\vec{b}) = \begin{bmatrix} 3s & 4 \\ -6 & 1 \end{bmatrix}$$

Computing $\det A$

$$\det A = 3s^2 - 12 = 3(s+2)(s-2)$$

The system has a unique solution precisely when $s \neq \pm 2$. For such an s , the solution is (x_1, x_2) , where

$$x_1 = \frac{\det A_1(\vec{b})}{\det A} = \frac{4s + 2}{3(s+2)(s-2)}$$

$$x_2 = \frac{\det A_2(\vec{b})}{\det A} = \frac{3s + 24}{3(s+2)(s-2)} = \frac{\cancel{3}(s+8)}{\cancel{3}(s+2)(s-2)}$$

3.3 Cramer's rule, Volume and Linear Transformations

IV

A formula for A^{-1}

Cramer's rule leads easily to a general formula for the inverse of an $n \times n$ matrix A . The j th column of A^{-1} is a vector $\vec{x}$ that satisfies

$$A\vec{x} = \vec{e}_j$$

where $\vec{e}_j$ is the j th column of the identity matrix, and the i th entry of $\vec{x}$ is the (i, j) -entry of A^{-1} . By Cramer's rule

$$\left\{ (i, j)\text{-entry of } A^{-1} \right\} = x_i = \frac{\det A_i(\vec{e}_j)}{\det A} \quad (2)$$

Recall that A_{ji} denotes the submatrix of A formed by deleting row j and column i . A cofactor expansion down column i of $A_i(\vec{e}_j)$ shows that

$$\det A_i(\vec{e}_j) = (-1)^{i+j} \det A_{ji} = C_{ji} \quad (3)$$

where C_{ji} is a cofactor of A .

By (2), the (i, j) -entry of A^{-1} is the cofactor C_{ji} divided by $\det A$. [Note that the subscripts on C_{ji} are the reverse of (i, j) .] Thus

$$A^{-1} = \frac{1}{\det A} \begin{bmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{bmatrix} \quad (4)$$

The matrix of cofactors on the right side of (4) is called the *adjugate* (or classical adjoint) of A , denoted $\text{adj } A$.

Theorem 8 An Inverse Formula

Let A be an invertible $n \times n$ matrix. Then

$$A^{-1} = \frac{1}{\det A} \text{adj } A$$

3.3 Cramers rule, Volume and Linear Transformations

V

Example 3

Find the inverse of the matrix $A = \begin{bmatrix} 2 & 1 & 3 \\ 1 & -1 & 1 \\ 1 & 4 & -2 \end{bmatrix}$

Solution

Let's start by computing the nine cofactors of A .

(remember the sign order)

$$C_{11} = + \begin{vmatrix} -1 & 1 \\ 4 & -2 \end{vmatrix} = -2 \quad C_{12} = - \begin{vmatrix} 1 & 1 \\ 1 & -2 \end{vmatrix} = 3 \quad C_{13} = + \begin{vmatrix} 1 & -1 \\ 1 & 4 \end{vmatrix} = 5$$

$$C_{21} = - \begin{vmatrix} 1 & 3 \\ 4 & -2 \end{vmatrix} = 14 \quad C_{22} = + \begin{vmatrix} 2 & 3 \\ 1 & -2 \end{vmatrix} = -7 \quad C_{23} = - \begin{vmatrix} 2 & 1 \\ 1 & 4 \end{vmatrix} = -7$$

$$C_{31} = + \begin{vmatrix} 1 & 3 \\ -1 & 1 \end{vmatrix} = 4 \quad C_{32} = - \begin{vmatrix} 2 & 3 \\ 1 & 1 \end{vmatrix} = 1 \quad C_{33} = + \begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix} = -3$$

The adjugate matrix is the transpose of the cofactor matrix.

$$\text{adj } A = \begin{bmatrix} -2 & 14 & 4 \\ 3 & -7 & 1 \\ 5 & -7 & -3 \end{bmatrix}$$

We could compute $\det A$ directly, but the following computation provides a check on the calculations for $\text{adj } A$ and produces $\det A$:

$$\text{adj } A \cdot A = \begin{bmatrix} -2 & 14 & 4 \\ 3 & -7 & 1 \\ 5 & -7 & -3 \end{bmatrix} \begin{bmatrix} 2 & 1 & 3 \\ 1 & -1 & 1 \\ 1 & 4 & -2 \end{bmatrix} = \begin{bmatrix} 14 & 0 & 0 \\ 0 & 14 & 0 \\ 0 & 0 & 14 \end{bmatrix} = 14I$$

Since $(\text{adj } A)A = 14I$, theorem 8 shows that $\det A = 14$.

3.3 Cramer's Rule, Volume and Linear Transformations (cont'd)

VI

Determinants as Area or Volume

In the next application, we verify the geometric interpretation of determinants described in the chapter introduction. Although a general discussion of length and distance in $\mathbb{R}^n$ will not be given until Chapter 6, we assume here that the usual Euclidean concepts of length, area, and volume are already understood for $\mathbb{R}^2$ and $\mathbb{R}^3$.

Theorem 9

If A is a 2×2 matrix, the area of the parallelogram determined by the columns of A is $|\det A|$. If A is a 3×3 matrix, the volume of the parallelepiped determined by the columns of A is $|\det A|$.

p.215

Proof 2×2

The theorem is obviously true for any 2×2 diagonal matrix

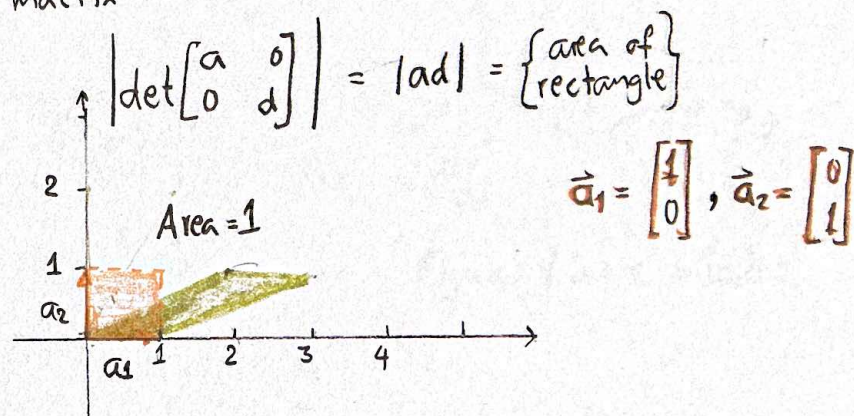


Figure 1: Shows determinant equals area

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, |\det A| = 1 = \text{area} = 1$$

Suppose we have $c=2$, and perform $2a_1 + a_2$

$$= \begin{bmatrix} 2 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\text{Our matrix } B = [a_1 \quad 2a_1 + a_2] = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

3.3 Cramer's rule, Volume and Linear Transformations (cont'd)

VII

Proof 3x3

The proof for $\mathbb{R}^3$ is similar. The theorem is obviously true for a 3×3 diagonal matrix, see figure.

Any 3×3 matrix can be transformed into a diagonal matrix using column operations that do not change $|\det A|$.

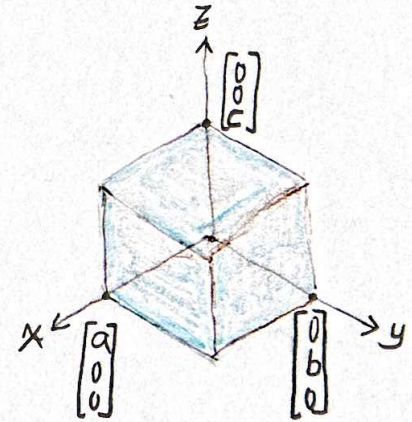


Figure: Volume = $|abc|$

Example 4

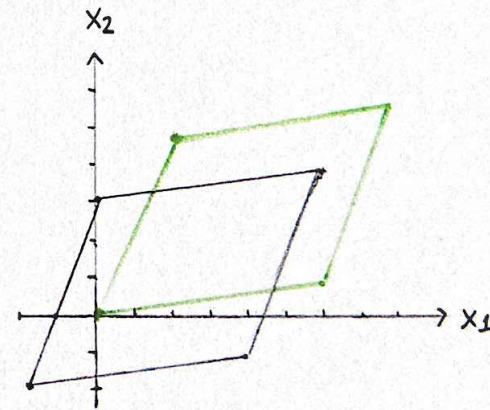
Calculate the area of the parallelogram determined by the points $(-2, -2)$, $(0, 3)$, $(4, -1)$, and $(6, 4)$

Solution

First translate the parallelogram to one having the origin as a vertex. For example, subtract the vertex $(-2, -2)$ from each of the four vertices. The new parallelogram has the same area, and its vertices are $(0, 0)$, $(2, 5)$, $(6, 1)$ and $(8, 6)$

The parallelogram is determined by the columns of

$$A = \begin{bmatrix} 2 & 6 \\ 5 & 1 \end{bmatrix} \quad |\det A| = |-28|, \text{ area} = 28.$$



3.3 Cramer's rule, Volume and Linear Transformations (cont'd)

VIII

Linear Transformations

Determinants can be used to describe an important geometric property of linear transformations in the plane and in $\mathbb{R}^3$. If T is a linear transformation and S is a set in the domain of T , let $T(S)$ denote the set of images of points in S . We are interested in how the area (or volume) of $T(S)$ compares with the area (or volume) of the original set S . For convenience, when S is a region bounded by a parallelogram, we also refer to S as a parallelogram.

Theorem 10

Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation determined by a 2×2 matrix A . If S is a parallelogram in $\mathbb{R}^2$, then

$$\{\text{area of } T(S)\} = |\det A| \cdot \{\text{area of } S\} \quad (5)$$

If T is determined by a 3×3 matrix A , and if S is a parallelepiped in $\mathbb{R}^3$, then

$$\{\text{volume of } T(S)\} = |\det A| \cdot \{\text{volume of } S\} \quad (6)$$

Proof

Consider the 2×2 case with $A = [\vec{a}_1 \ \vec{a}_2]$. A parallelogram at the origin in $\mathbb{R}^2$ determined by vectors $\vec{b}_1$ and $\vec{b}_2$ has the form

$$S = \{s_1 \vec{b}_1 + s_2 \vec{b}_2 : 0 \leq s_1 \leq 1, 0 \leq s_2 \leq 1\}$$

The image of S under T consists of points of the form

$$\begin{aligned} T(s_1 \vec{b}_1 + s_2 \vec{b}_2) &= s_1 T(\vec{b}_1) + s_2 T(\vec{b}_2) \\ &= s_1 A \vec{b}_1 + s_2 A \vec{b}_2 \end{aligned}$$

where $0 \leq s_1 \leq 1, 0 \leq s_2 \leq 1$. It follows that $T(S)$ is the parallelogram determined by the columns of the matrix $[A \vec{b}_1 \ A \vec{b}_2]$. This matrix can be written as AB , where $B = [\vec{b}_1 \ \vec{b}_2]$. By theorem 9 and the product theorem for determinants,

$$\begin{aligned} \{\text{area of } T(S)\} &= |\det AB| = |\det A| |\det B| \\ &= |\det A| \cdot \{\text{area of } S\} \end{aligned} \quad (7)$$

%

3.3 Cramer's rule, Volume and Linear Transformations (cont'd)

IX

An arbitrary parallelogram has the form $\vec{p} + S$, where $\vec{p}$ is a vector and S is a parallelogram at the origin, as seen previously. It is easy to see that T transforms $\vec{p} + S$ into $T(\vec{p}) + T(S)$. Since translation does not affect the area of a set

$$\begin{aligned}\{\text{area of } T(\vec{p} + S)\} &= \{\text{area of } T(\vec{p}) + T(S)\} \\ &= \{\text{area of } T(S)\} \quad (\text{translation}) \\ &= |\det A| \cdot \{\text{area of } S\} \quad (\text{By equation (7)}) \\ &= |\det A| \cdot \{\text{area of } (\vec{p} + S)\} \quad (\text{translation})\end{aligned}$$

This shows that (5) holds for all parallelograms in $\mathbb{R}^2$. The proof of (6) for the 3×3 case is analogous. ■

Example 5

Let a and b be positive numbers. Find the area of the region E bounded by the ellipse whose equation

is

$$\frac{x_1^2}{a^2} + \frac{x_2^2}{b^2} = 1$$

Solution

We claim that E is the image of the unit disc D under the linear transformation T determined by the matrix $A = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$, because if $\vec{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$, $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, and $\vec{x} = A\vec{u}$, then

$$u_1 = \frac{x_1}{a} \quad \text{and} \quad u_2 = \frac{x_2}{b}$$

It follows that $\vec{u}$ is in the unit disk, with $u_1^2 + u_2^2 \leq 1$, if and only if $\vec{x}$ is in E , with

$\left(\frac{x_1}{a}\right)^2 + \left(\frac{x_2}{b}\right)^2 \leq 1$. By the generalization of theorem 10,

$$\begin{aligned}\{\text{area of ellipse}\} &= \{\text{area of } T(D)\} \\ &= |\det A| \cdot \{\text{area of } D\} \\ &= ab\pi(1)^2 = \pi ab\end{aligned}$$
 ■